

Promoting Inclusive Open Education: A Holistic Approach Towards a Novel Accessible OER Recommender System

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Abstract. OER can pave the way to reach equitable and inclusive access to education for all. However, unleashing its full potential is disrupted due to the disregard of the special needs and learning characteristics of certain user groups such as people with disabilities by the mechanisms delivering OER. A potential solution lies within helping each final user to find the "best" OER that embraces his/her individual needs and specific characteristics. Recommendation accomplishes that as it could offer a significant support assisting learners with disabilities in recognizing and retrieving the most relevant accessible OER within the system delivering the educational content. This paper lays the foundation to an approach for building an accessible OER recommender system that delivers, to online learners, the most relevant accessible OER responsive to their needs, abilities, and disabilities, based on their contextual and usage data.

Keywords: Open Educational Resources · OER · Accessibility · Recommendation systems · People with disabilities · PWD · OER repositories · Inclusive education · Open education

1 Introduction

According to the United Nations 2030 agenda for sustainable Development, "The spread of information and communication technologies and global interconnectedness has great potential to accelerate human progress, to bridge the digital divide and to develop knowledge societies". This vision spurred the inception of Open Educational Resources (OER) movement which aims to remove all barriers to education [18]. OER refer to teaching, learning, and research materials that are either in the public domain or available, under an open license, for free access, use, adaptation, and redistribution by others with no or limited restrictions [33]. OER hold the promise of inclusive and

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equitable access to education through enabling, via its transformative potential, educational content to be adapted to fit the specific needs of people with disabilities [38]. People with disabilities are vulnerable to the deficiencies in educational opportunities recognizing their particular needs. This reinforces their exclusion especially from formal education, resulting in unemployment, poverty, inequality, and instability [14, 33]. To that extent, the OER paradigm fosters inclusive education through holding a transformative potential supporting the adjustments to be performed on the educational content and adding more flexibility to the educational environment [7, 14]. Despite that several research focused on OER, little attention was paid to OER accessibility. Learners with disabilities are still facing the "non" accessible educational content. This is induced by the fact that OER producers seem oblivious to the importance of adopting accessibility requirements when creating OER, and the non-support of the adjustments that need to be done within learning environments and OER delivery systems, that are intended to facilitate the search, use and exploitation of OER for people with disabilities [2]. Confirming that, an analysis performed in [2] exposed two main barriers impeding open inclusive access to OER for all users. The first barrier concerns the non-consideration of personalization mechanisms as an OER delivery system should enable each learner – in particular those with disabilities – to locate and retrieve educational content matching his/her learning needs, abilities/disabilities, and individual characteristics. While the second barrier is related to a non-effective or a total discard of accessibility standards, showing that most of existing OER delivery systems do not involve people with disabilities. In short, existing OER delivery systems provide generally the same content, mostly not in compliance with accessibility principles, to all learners, not considering that they learn with not only different needs but also different abilities, disabilities, and learning characteristics. Which deepens the digital divide and exclusion from educational opportunities [2].

A potential way to balance this out is enacting personalization paradigms such as recommendation systems. These systems hold the promise to help and assist users, especially those living with a disability, to easily find and retrieve the most relevant educational content responding to their needs and interests [4, 24]. In this context, this study presents a new approach instrumentalizing the general recommendation paradigm towards recommending personalized accessible OER within OER delivery systems for people with disabilities. Additionally, the proposed approach aims to explore and exploit the growing number of existing non-accessible OER through enabling their conversion before providing learners with the most suitable and accessible personalized OER recommendations.

To this end, this paper unfolds 4 sections. After the introduction, Sect. 2 gives an overview of the scope of this work covering OER and accessibility and their related concepts. Section 3 presents the recommendation paradigm in education and accessibility. Section 4 finalizes the paper through a conclusion.

2 Open Inclusive Education

2.1 Accessibility

Through its world report on disability, WHO[1] (World Health Organization) revealed that over a billion people live with a form of disability. However, despite their significant number people with disabilities are one of the most marginalized groups in the world [36]. They endure difficulties in daily life aspects emphasized by the "non" adaptation of services, products or environments to their particular needs and specific characteristics. The accessibility paradigm counters these difficulties through providing concrete recommendations and practices to enable people with disabilities to achieve, on equal basis with others, access to education, employment, transportation, housing, etc.) [14]. Accessibility grants the "ability to access" to devices, services, products or environments for people living with a disability through the use of assistive technologies [15]. In line with that, Microsoft scaffolded that "There are no limits to what people can achieve when technology reflects the diversity of everyone who uses it" [20] proving that enabling equal access to technology especially for people with disabilities provides them with limitless opportunities to actively participate in society and empower them to overcome the difficulties they face in everyday life.

Falling in the scope of disability rights movement [14], the 9th article of United Nations Convention on the Rights of Persons with Disabilities consolidated this idea through advocating the crucial importance of universal access to technology especially the web. In this scope, accessibility in general, and web accessibility in particular, rhymes with democratizing access to Information and Communication Technologies (ICT) for People With Disabilities PWD through promoting the design, development, production and distribution of accessible web technologies. Web accessibility fosters the ability of people with disabilities to use and contribute to the web [14]. To this end, the Web Accessibility Initiative (WAI), which is a wing of the World Wide Web Consortium (W3C), is established and aims to lead the Web to its full potential to be accessible, enabling people with disabilities to participate equally on the Web. The Web Accessibility Initiative (WAI) "develops strategies, guidelines, and resources to help making the Web accessible to people with disabilities" [14]. The Web Accessibility Initiative (WAI) encompasses guidelines and recommended practices regarding the accessibility of: content through the Web Content Accessibility Guidelines (WCAG), authoring tools through Authoring Tools Accessibility Guidelines (ATAG) and user agents through User Agent Accessibility Guidelines (UAAG) [14].

Through its 26th article, the UN Convention on the Rights of Persons with Disabilities recognizes education as a fundamental human right highlighting its importance in everyone's sustainable personal development. People with disabilities are vulnerable to the deficiencies in educational opportunities recognizing their particular needs. This reinforces their exclusion from education which generates skill gaps and causes unemployment, poverty, inequality, and instability [37]. In education, accessibility promotes flexible and personalized learning for PWD through fostering the design and

[1] http://www.who.int/en/.

development of educational content, activities and services meeting their requirements and characteristics. In the same scope, given a re-definition of the term disability by the IMS Global Learning Consortium describing it as a mismatch between the needs of the learner and the education offered, accessibility refers to the ability of the learning environment to adjust to the needs of all learners. It is determined by the flexibility of the educational environment and the availability of adequate alternative-but-equivalent content and activities [14].

In this context, the OER paradigm fosters inclusive education through holding a transformative potential supporting the adjustments to be performed on the educational content and adding more flexibility to the educational environment [8].

2.2 Open Educational Resources (OER)

The urging need to open up, share freely and give wider access to information and technology, has motivated a range of open philosophies such as: Open source, Open access, Open Content, Open Data, Open Education, etc. The OER concept falls in the scope of the Open Education movement covering cost and constraint free resources, tools and practices that foster open sharing to broaden access to education [19]. The emergence of this concept encouraged and spurred the educational institutions to work on making education available for all. The Massachusetts Institute of Technology (MIT) was one of the first institutions to join the 'Open' wave through making nearly all its courseware freely available for all in 2001. Later, the number of institutions offering free or open courseware kept increasing. That is when UNESCO organized the 1st Global OER Forum in 2002 where the term Open Educational Resources (OER) was first coined and presented as: *"The open provision of educational resources, enabled by information and communication technologies, for consultation, use and adaptation by a community of users for non-commercial purposes"*. Then and with the constant growth of the OER movement, other efforts were made to define and describe the term and the concept of OER. In the Cape Town Open Education Declaration, during the conference on open education in 2007, OER were defined as follows "*open educational resources should be freely shared through open licenses which facilitate use, revision, translation, improvement and sharing by anyone*."

However, it was not formally adopted until 2012 through the Paris OER Declaration [31] which defined OER as: "*teaching, learning and research materials in any medium, digital or otherwise, that reside in the public domain or have been released under an open license that permits no-cost access, use, adaptation and redistribution by others with no or limited restrictions. Open licensing is built within the existing framework of intellectual property rights as defined by relevant international conventions and respects the authorship of the work"*.

Keeping in mind the aforementioned definitions, OER are cost-and-constraint (or limited constraints) free, flexible educational materials for all. OER are intended to foster and promote educational transformation. Besides, the most powerful asset of open educational resources is its ability, when digitalized, to be easily accessed, used, re-used and shared online.

Both OER and Accessibility are important pillars holding great potential to effectively build inclusive knowledge societies through removing the barriers that may hold back the expansion of education for all especially people with disabilities.

2.3 Accessible OER

Stemming from recognizing accessibility as a main design requirement when creating educational content, several research initiatives investigated the potentials of providing accessible educational content tailored to learners with disabilities' needs and characteristics. These initiatives gave life to a number of systems aiming to foreclose digital exclusion such as: The EU4ALL project [9], MoodleAcc+ platform [17], etc.

Following the same line and along with the tidal wave of designing for inclusion, researches took further the efforts towards maximizing flexibility and customization capabilities of the educational content to meet the individual needs of each and every learner. Which led to developing educational content as Open Educational Resources [10] fulfilling the promise of not only flexibility and transformability but also ruled out several financial, technical and access barriers.

As OER represent an important milestone to achieve equitable inclusive education, UNESCO[2] was proactive releasing OER-based inclusive guidelines and recommendations to promote and empower equitable inclusive access to education. The most important are:

- The "Guidelines for OER in higher Education"[3] in collaboration with CoL (Commonwealth of Learning) which highlighted, on one hand, that OER should be open and accessible to students with a diversity of learning needs. On the other hand, the guidelines pinpointed that the latter is achieved through the resource delivery system which will handle any "transformation" the educational resource might require to match the individual learning needs of the students.
- The Ljubljana OER Action Plan 2017, released as an outcome of the 2nd OER world congress and which described concrete actions to develop and use accessible inclusive OER. The suggested actions shed the light on the need to acknowledge the principle of inclusiveness when creating, accessing and sharing OER [21].
- And last but not least, UNESCO has recently furthered its efforts through releasing a draft recommendation on OER [7] in which it presents a specific list of important measures to scaffold the creation, use and adaptation of inclusive and quality OER [7].

Thus, two main facts can be revealed: One recognizing that increasing accessibility is crucial to reach inclusive knowledge society and the other proving that the latter can only be achieved through a close implication of OER creators and OER delivery systems.

[2] https://en.unesco.org.

[3] https://unesdoc.unesco.org/ark:/48223/pf0000213605.

2.4 OER Delivery Systems

To find OER, the user needs to explore the different kinds of facilities and services collecting, storing and supplying user communities with OER. These facilities support web-based management of OER through easing the search, retrieval and access to learning resources [26, 27]. Underpinning that, Sampson et al. [26, 27] assert that the main aim of OER delivery systems is to manage the use, organization, classification and share of OER in the form of Learning Objects (LO) and their associated metadata [26].

There is no shortage of optimism about the potential of OER in reducing barriers to education and improving the learning quality of PWD. However, a serious engagement from the mechanisms delivering OER is crucial to bring out and explore this potential through the functionalities and access features they offer. That's why there's a dire need to pursue a close investigation of the OER delivery systems commitment degree to OER main goals for providing broader, more equal and barrier free access to knowledge and education for all users.

Within this scope, a close analysis of a representative sample of twenty existing OER delivery systems in [2] exposed two main drawbacks; the first covering the non-consideration of personalization since OER delivery systems should enable each learner– in particular those with disabilities – to locate and retrieve educational content matching his/her learning needs and individual characteristics. While the second revealed a non-effective or a total discard of accessibility standards showing that most of these OER delivery systems do not involve people with disabilities through delivering the same content to all learners not considering that students learn with not only different needs but also different abilities, disabilities, and learning characteristics. Which deepens the digital divide and exclusion from educational opportunities [2].

A potential avenue to counter the lack of personalization for PWD in OER delivery systems is considering content discovery mechanisms like recommender systems. These systems can provide a significant support guiding learners with disabilities through locating as well as retrieving the most relevant accessible open educational resources meeting their disabilities and learning objectives. A special focus should be oriented consequently towards investigating the challenges of deploying recommendation services to foster people with disabilities' learning within OER delivery systems.

3 Accessible OER Recommendation

3.1 OER Recommendation and Accessibility

One of the most recurrent dilemmas facing users searching for educational content is the tremendous number of learning resources available on the internet which makes finding the content they need more difficult. Unwin, in [34], confirms that the real issue is not the lack of content but the inability of end users to effectively locate relevant learning resources [34]. Recommender systems emerge like the answer most likely to address this issue. The main purpose of these systems is to help users through pre-selecting resources that respond to their needs, interests and objectives [16]. Confirming that, Burke, R. [4] states that a recommender system is any system that

produces personalized recommendations as output or has the effect of guiding the user in a personalized way to interesting or useful objects in a large space of possible options. In education, personalized recommender systems assist and support learners in achieving an optimal personalized learning experience through their ability to anticipate learners' needs and to provide them with matching information and content. According to Resnick and Varian [24], recommendation in learning environments relies on using the experience of a community of learners to help individual learners in that community to identify and retrieve more effectively learning content from a wide spectrum of choices.

Recommenders can be applied to a variety of web based learning environments such as OER systems [16]. In that context, embodying recommenders into OER-based learning environment federates the personalization and adaptation features offered by the recommendation paradigm along with the transformative potential of OER. This would pave the way to OER-based personalized learning experiences for learners with disabilities recognizing their evolving and special needs, and taking into account the wide range of their accessibility requirements and preferences. Which represents a step forward towards equitable inclusive education.

3.2 Approach Towards a Novel Accessible OER Recommendation System

The proposed approach was built upon the Generic Meta-Level GML Framework for Technology Enhanced Learning (TEL) Recommendation Systems described in [16]. This approach (see Fig. 1) follows several steps beginning with retrieving, federating then pre-processing learners' and content data to build learner and OER models, to be used later within the recommendation engine. Afterward the OER conversion process, whenever required, is launched to perform the conversion of appropriate recommended OER into accessible ones responding to the active learner profile and her contextual characteristics. The output of the recommendation engine is then delivered to the learner through a specific accessibility viewer/delivery tool acknowledging his output device considering the potential use of assistive technologies.

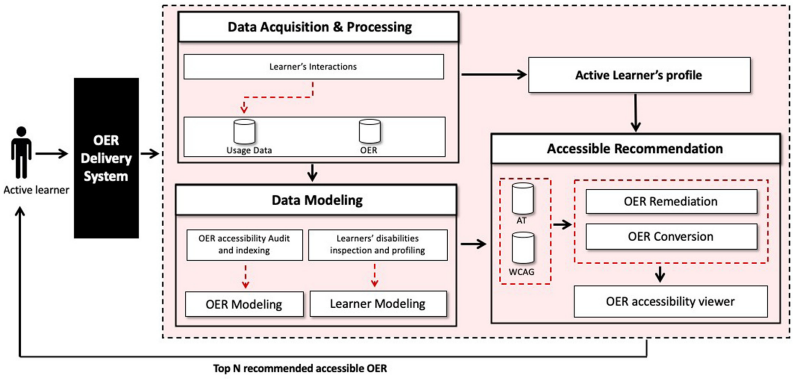


Fig. 1. Proposed accessible recommendation approach.

As described in Fig. 1, the proposed approach involves the following primary phases:

- **Data acquisition and processing:** This phase handles acquiring and mining the data describing the learners' characteristics and usage data (learners' usage activities, learners' disabilities and characteristics, contextual data such as space-time features (location, luminosity, sound level, time, etc.), used device, etc. [1, 35], assistive technologies' usage, etc.) and the educational content (OER and the metadata). It represents the system's learning phase handling the acquisition, retrieving, filtering and exploitation of data towards building learner and content profiles.
- **Modeling phase:** This phase is performed offline. It covers the modeling processes of learners and OER. (1) Learner modeling: The previously processed data is used to build learner models through the use of machine learning algorithms. Thereafter, learner models are grouped into several collections gathering similar learners with very much alike characteristics and preferences, to be used later on in the recommendation phase. (2) OER modeling: this process handles the automated exploration, crawling, analysis and indexing of OER features and content toward building an entire index representing all available resources in the repository taking into account accessibility issues. In order to consider the accessibility aspect of OER modeling, we include an offline auditing phase of OER to check and inspect their compliance with known accessible requirements and then generating a specific report explaining results and stating possible curation and remediation. This output is used either for modeling and conversion phases.
- **Accessibility tasks:** It represent tasks related to accessibility issues and covering mainly services of auditing, converting and viewing accessible OER. It encompasses the following tasks: (1) OER auditing: This process is performed offline and manages the accessibility evaluation, checking and testing of OER available in the repository, verifying thus their compliance with the WAI-WCAG requirements. Reporting templates are then used to report the evaluation results which will be processed with the OER features to build the OER model. (2) OER Conversion: covers the on-demand conversion of the "non" accessible OER. The conversion applies, whenever it is possible, the remediation options and strategies referring to the techniques addressing the learning resource's accessibility flaws. An automatic update of the content model as well as the OER repository is then performed. (3) OER viewing: it aims to display the recommended accessible OER with respect to the features of the assistive technologies used by the learner and its output device.
- **Recommendation phase:** This will ensure the prediction of the learning content, anticipating the learners needs and disability characteristics. In fact, it needs to be preceded by the identification of the active learner's profile through extracting the relevant information characterizing him/her. Once the active profile is recognized, the recommendation process is launched, and a prediction list of suitable OER is generated based on the learner's needs and disability characteristics. If not already accessible and appropriate to be used by the learner, the recommended OER is then converted-whenever it is possible, into accessible OER based on the remediation options and strategies provided as a result to the OER audit task. The top

N recommended accessible OER are then provided to the active learner through the accessibility viewer in a format compliant to the output device and assistive technologies he/she used.

- **Active learner profile identification:** The active learner refers to the current online learner using the OER delivery system. Identifying his/her profile is essential to guarantee the accuracy of the prediction process. It ensures recognizing the present request, needs and characteristics of the learner requiring recommendation. Through the active learner's interactions with the OER delivery system, his/her relative activities and contextual data, the assistive technologies' usage and the disability characteristics are identified (in an implicit and/or explicit manner) and extracted, processed and delivered to the recommendation engine toward providing the active learner with appropriate content matching his/her preferences and disabilities' requirements.

4 Conclusion

This paper gave an overview of a proposed approach towards recommending personalized accessible Open Educational Resources. We highlighted the core modules and functionalities leading the proposed approach. Our future works involve furthering our studies of each module and each step of the targeted recommendation system to explore and select most suitable techniques and algorithms to be used within each phase, which will pave the way to the development of a prototype of an accessible OER recommendation service to be deployed and experienced within existing OER delivery systems.

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